PART B – BASELINE UNIT TESTING SUITE

Test ID	Test Case	Expected Result	Legacy Baseline	Refactored Code Result
TC-01	Add 10, 20, 30	All three items are stored/displayed correctly.	PASS	PASS – addItem stores all three values.
TC-02	Add 6 items to trigger resize	Resize occurs and data remains intact.	FAIL – old heap is leaked.	PASS – resize copies data and delete[] releases old storage.
TC-03	Print 5 items	Exactly 5 valid items are printed.	FAIL – garbage/out-of-bounds read.	PASS – printAll uses i < size.
TC-04	Find 30	Returns index 2.	PASS – index 2.	PASS – findItem returns index 2.
TC-05	Find 999	Returns -1.	PASS – -1.	PASS – returns -1 when not found.
TC-06	Remove index 2	Item at index 2 is removed and later items shift left.	PASS	PASS – valid index is removed and remaining items shift once.
TC-07	Remove index -5	Invalid input is rejected safely.	PASS	PASS – index < 0 is rejected.
TC-08	Remove index 99	Invalid input is rejected safely.	FAIL – missing upper-bound validation.	PASS – index >= size is rejected.
TC-09	Transpose 3×3 matrix	Rows and columns are transposed correctly.	PASS	PASS – same 3×3 transpose logic is retained.
TC-10	Program exit / cleanup	All dynamically allocated memory is released.	FAIL – final leak.	PASS – DynamicArray destructor releases data automatically.